

An attentional-bias account of language emergence

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Abstract

Why do humans alone develop language? I argue that evolution selected a binding bias, not specialized syntactic machinery: a hyperprior treating addressed, multi-timescale communicative events (infant-directed speech, with eye contact and rhythmic prosody) as reliable evidence. Under active inference, the same precision profile biases both perceptual uptake and vocal convergence; no separate imitation system is required. Human echolocation illustrates the logic: trained blind individuals navigate by self-generated clicks, a complex skill from generic neural systems; complexity doesn't entail selection. The account explains language's universality and the combinatorial limits of other vocal learners. Three falsifiable tests: a neonatal preference study, a computational replay varying the gain profile, and a newborn functional near-infrared spectroscopy (fNIRS) assay. Selection targeted what we notice, not what we compute.

Keywords: language emergence, attention bias, echolocation, spandrel, gene–culture coevolution, hyperprior

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I INTRODUCTION

Why do humans alone among living species develop language? The question has driven decades of debate between two camps. One posits specialized neural architecture that evolved for language (Berwick & Chomsky, 2016; Chomsky, 1965; Fitch et al., 2005; Hauser et al., 2002). The other attributes acquisition to domain-general mechanisms shaped by culture (Christiansen & Chater, 2015; Evans & Levinson, 2009; Scholz et al., 2025). Recent reviews integrate these perspectives within a biocultural framework (Arnon et al., 2025); the present account proposes a concrete mechanism (attentional precision-weighting) that could anchor that integration.

Proponents of the first view invoke computational-complexity and poverty-of-stimulus (PoS) arguments, concluding that humans have to possess innate, language-specific modules (Baker, 1979; Chomsky, 1980). But child-scale sequence models already distinguish licit from illicit grammars, sharply undercutting PoS claims (Kallini et al., 2024; Perfors et al., 2011; Warstadt et al., 2023). Cultural-evolution accounts, for their part, show how languages adapt to cognitive constraints through transmission (Kirby et al., 2008), which in turn bootstraps acquisition (Thompson et al., 2016). Still, neither camp explains the deeper puzzle: why humans tune in while every other species tunes out (cf. Scott-Phillips, 2017).

I propose the ATTENTION-BIAS HYPOTHESIS: humans inherit hyperpriors that boost precision on the bound event of ostensive, multi-timescale communicative interaction (e.g. infant-directed speech with addressed gaze). By treating the addressed communicative event as high-precision evidence, the bias steers domain-general learning toward language without specialized computation, rich innate knowledge, or elaborate evolutionary hurdles; a structured gain profile, combined with rich communicative input, can support a learning trajectory toward linguistic structure.

Converging evidence challenges dedicated-faculty accounts. Domain-general action-sequence mechanisms plausibly support language (Kolodny & Edelman, 2018); generic learners in computational models master human languages but stumble on unattested ones (Kallini et al., 2024; Warstadt et al., 2020). Such results suggest that combinatorial linguistic capacity emerges from exapted, domain-general neural stock rather than direct selection for a language-specific processor (S. J. Gould & Lewontin, 1979). What selection *did* target, on the present account, is the attentional gain mechanism that makes the capacity reliably expressed.

The spandrel view gains unexpected support from human echolocation. Specialized-faculty advocates often invoke bat echolocation: bats genetically evolved for ultrasonic navigation, so humans must have done the same for language by the same logic. The analogy collapses once we see that humans echolocate too. Blind individuals, despite

no specialized adaptations, navigate with self-generated clicks and achieve spatial resolutions rivaling bats (Dodsworth et al., 2020; Teng & Whitney, 2011). Complexity alone need not imply genetic adaptation. Because humans lack an innate bias toward echo patterns, echolocation remains latent in nearly all sighted individuals, underscoring attention's central role.

Language and echolocation are both complex achievements attainable with domain-general machinery. Evolution tipped the scales only for language, endowing infants with an attentional bias toward addressed communicative events. We all acquire language because we can't avoid noticing it; we rarely echolocate because nothing compels us to attend.

This paper develops the attention-bias account in five stages. Section 2 treats human echolocation as a spandrel-like latent capacity (a complex ability that emerges from generic auditory-spatial machinery without evidence of direct selection for echo-navigation), which undermines arguments for specialized language faculties from complexity alone. Section 3 gathers evidence for innate attentional biases and formalizes precision-weighting as the mechanism. Section 4 outlines three empirical tests: cross-species preference studies, computational models that vary the gain profile, and neonatal neuroimaging of precision-weighting. Section 5 draws implications for language emergence within gene-culture coevolution, and Section 6 closes. The account reframes language evolution as selection for what we notice, not what we compute.

2 ECHOLOCATION: SPANDRELS, ADAPTATIONS, AND EVOLUTIONARY INSIGHT

2.1 KEY EVOLUTIONARY TERMS

Following Gould and colleagues, I treat three labels as analytically distinct. An ADAPTATION is a trait moulded by natural selection for the function it now serves; the claim requires evidence that historical selection targeted that function and that the trait's structure shows design for it (S. J. Gould & Vrba, 1982). An EXAPTATION is a trait that performs a current role but originated either under selection for a different role or as a neutral architectural by-product. A SPANDREL is the limiting case: a feature that arises as a structural by-product of adjoining adaptations and was never itself a target of selection (S. J. Gould & Lewontin, 1979). I use EXAPTATION when the evidence supports co-option of pre-existing capacity, and SPANDREL more loosely for capacities that emerge from domain-general neural stock without any prior adaptive history for that function.

Diagnosing which category applies demands converging evidence. Specialized peripheral anatomy argues for adaptation; generic hardware that is merely pressed into new service supports an exaptation or spandrel reading. Dedicated neural circuitry also counts as structural specialization, even in the absence of distinctive anatomy, whereas flexible

computational implementation suggests exaptation. Developmental data provide a further clue: a capacity that emerges reliably without tuition is more plausibly an adaptation, while one that appears only with explicit training is better viewed as a spandrel.

2.2 ECHOLOCATION ACROSS SPECIES

Echolocation ranges from rudimentary obstacle detection to millimetre-scale prey tracking. In echolocating bats and odontocetes, the capacity is unmistakably an adaptation. Both groups display peripheral specializations: enlarged cochleae, stiffened basilar membranes, and the convergently evolved motor protein *prestin* in outer hair cells (Dawkins, 2024; Vater et al., 2004). Bat pups begin emitting task-ready pulse trains on their first flights (19–20 d postnatal) without tutelage (Buchler, 1980), and adults resolve prey size, texture, and movement with striking precision (Wilkinson, 2003). Neural pathways from inferior colliculus to auditory cortex are dedicated to echo delay and Doppler analysis, yielding real-time spatial maps (Aihara et al., 2013). Together these anatomical, developmental, and functional lines leave little doubt that ultrasonic echolocation is an adaptation.¹

Humans present a contrasting picture. The auditory periphery is ordinary, operating in the 2–10 kHz band; no structural tweaks hint at selection for echo processing. Nonetheless, both blind and sighted adults can be trained to extract detailed spatial information from mouth-click echoes (Dodsworth et al., 2020; Kolarik et al., 2014). Expert blind echolocators discriminate horizontal offsets of about 1.2°, approaching the limits of human binaural localization and comparable to bat performance on matched tasks (Teng & Whitney, 2011, p. 483). Surveys put uptake among congenitally blind adults near 20–30 %, likely conservative because many users employ the skill tacitly (Thaler, 2013). Latent motor abilities of this sort aren't unusual: most people neglect the auricular muscles despite having the hardware (Bérzin & Fortinguerra, 1993).

The neural implementation diverges sharply from the bat template. Functional imaging shows that blind echolocators commandeer occipital cortex. Primary visual areas develop coarse retinotopic-like maps for echo delay (Norman & Thaler, 2019), the occipital place area tracks echo-defined routes (Norman & Thaler, 2023), and parahippocampal cortex codes surface material from echoes (Milne et al., 2015). Short-term training in sighted adults induces parallel shifts, confirming that these changes are experience driven rather than hard-wired (Norman et al., 2024). The absence of peripheral specialization, the need for explicit practice, and the repurposing of visual regions together identify human echolocation as a spandrel, not an adaptation.

¹Putnam's 1980, p. 405 claim that roughly seven-eighths of the bat brain is auditory is off by an order of magnitude: quantitative volumetric studies place the major auditory nuclei far below that fraction, even in insectivorous echolocators (Fortin et al., 2008; Hutcheon et al., 2002; Rademacher et al., 2001; Rombaux et al., 2010).

Cross-modal plasticity shows that human echolocation relies on repurposing existing neural systems rather than dedicated circuits evolved specifically for echolocation. Table 1 summarizes the key neuro-imaging evidence.

Table 1: Neural correlates of human echolocation

Study	Participants	Key finding	Effect metric
Norman & Thaler (2019)	5 blind experts; 10 sighted controls	Retinotopic-like spatial maps for echoes in primary visual cortex	$t_{10} = 3.675$; $d \approx 2.0$ (95 % confidence interval (CI) 0.7–3.3) ^a
Norman & Thaler (2023)	6 blind experts; 12 blind controls; 14 sighted controls	Spatial-route echoes elicit stronger occipital-place-area activation than scrambled echoes	Experts vs. sighted: $d \approx 1.5$ ^b
Norman <i>et al.</i> (2024)	26 adults (blind + sighted) pre/post training	Echo training reshapes left-hemisphere primary visual cortex (V1) representation	Within-subject $F_{1,24} = 4.948$; $d_z \approx 0.44$ ^c
Milne <i>et al.</i> (2015)	3 blind experts; 6 blind controls	Parahippocampal cortex tracks echo-based material judgements	Behavioural four-alternative forced-choice (4-AFC) gap $d \approx 3.4$ ^d

^a $d = t\sqrt{1/n_1 + 1/n_2}$; CI via Hedges' g .

^b From published means and pooled standard deviation (SD).

^c $d_z = t/\sqrt{N}$.

^d Imaging single-subject; behavioural statistic shown.

2.3 HUMAN ECHOLOCATION AS A SPANDREL

Because most sighted humans never need, notice, or practise echolocation, the trait is unlikely to have been a sustained target of selection. Its hallmarks are those of a spandrel: no peripheral specialization, reliance on neural circuits built for other tasks, and emergence only under explicit training or sensory loss.

The cortical machinery comes largely from vision. Echo cues recruit early visual areas, the occipital place area, and parahippocampal regions, territory originally tuned for spatial layout and surface properties. The auditory periphery remains generic, and behavioural onset is experience dependent, in sharp contrast to the automatic, pup-onset system of bats. Human echolocation is neither rudimentary nor ersatz; its performance ceiling rivals bat localization once practice drives cortical re-allocation. What differs is evolutionary provenance, not computational sophistication.

Latent audible-frequency echolocation also appears in shrews and rats (E. Gould *et al.*, 1964; Rosenzweig *et al.*, 1955), reinforcing the point that echo processing can hitch-hike on general auditory and spatial circuits. Dormant capacities of this sort are not unique to hearing. In vision, roughly 12 % of women carry a fourth cone pigment

but seldom experience novel colour categories (Jordan et al., 2010). Both cases illustrate how complex capacities can sit unused until circumstances or training expose them.

Human echolocation thus exemplifies a broader principle: plastic neural architecture can yield high-level abilities without prior adaptive design. The capacity is real, potent, and biologically cheap, precisely the profile expected of a spandrel.

2.4 IMPLICATIONS FOR LANGUAGE-EMERGENCE ARGUMENTS

Adaptationist defences of a language-specific instinct often lean on an analogy to bat echolocation (Fitch, 2014; Jackendoff, 2002; Pinker & Bloom, 1990; Pinker & Jackendoff, 2005; Tooby et al., 2005). The argument may be spelled out as:

- Premise 1: Bat ultrasonic echolocation is a complex, species-typical capacity.
- Premise 2: It arose through direct selection for that function.
- Premise 3: Human language is likewise complex and species-typical.
- Bridge: Any complex, species-typical capacity has to arise through direct selection for its current role.
- Conclusion: Language is a direct adaptation.

Latent human echolocation breaks the bridge. The ability is complex and widely attainable, but it (i) typically requires deliberate training, (ii) uses unspecialized auditory hardware, and (iii) shows no phylogenetic trace of selection for echo-navigation. On the present usage it's a spandrel and a counter-example to the inference "complex → adaptation." The echolocation template offers no evidential support for a language-specific module; such a claim stands on genetic, anatomical, or fitness data of its own. The live evolutionary question becomes: which inherited biases draw human infants – but not other species – to structured vocal–gestural input (Vernes, 2017)?

2.5 ADDRESSING THE UNIVERSALITY ASYMMETRY

The spandrel reading invites a likely objection: the distributional gap. Language emerges in every neurotypical child without instruction, whereas echolocation surfaces only in a minority of blind adults. If both capacities are spandrels, why the asymmetry?

The *attention-bias hypothesis* offers a direct answer: selection, I contend, targeted not the evolution of specialized language processors, but the tuning of an inherited gain mechanism that assigns high precision to ostensive, multi-timescale communicative events. That boost magnifies linguistic prediction errors, making them the most informative cues for a generic learner (see Section 3). No comparable bias exists in humans for click echoes, so the latent skill remains dormant unless blindness or formal training forces attention onto those cues.

A self-domestication route is one evolutionary scenario in which the gating circuitry behind this profile could be retuned: selection against reactive aggression, with downstream effects on stress reactivity and social tolerance, could shift how subcortical and

basal-ganglia circuits weight salience and reward for social cues (Wilkins et al., 2014; Wrangham & Hare, 2019). The argument here doesn't depend on any specific mechanistic chain. What matters is architectural: regulatory tuning of the circuitry that gates which streams receive precision is the kind of change a gain-policy account would predict, whatever the developmental route. The Casten et al. (2026) findings on human ancestor quickly evolved regions (HAQERs), discussed in §5, fit this picture, with regulatory regions enriched near striatal medium spiny neurons and *FOXP2*-expressing neurons carrying a polygenic signal associated with spoken-language phenotypes. Because the HAQER signal is polygenic, balancing-selected, and pre-Neanderthal in origin (Casten et al., 2026), it points to a deeper proto-hominin regulatory substrate on which any later sapiens-scale self-domestication would have acted, rather than to a single recent dosage-sensitive locus.

Without an attentional gain boost, echolocation stays rare; with it, language becomes universal. The empirical test of this asymmetry is the cross-species neonatal preference study in §4.1: if non-human neonates show the same preference ordering under species-appropriate ostensive/rhythmic stimuli, the human-specificity claim is weakened. The contrast preserves the lesson of the analogy: complex skills can crystallize from domain-general neural stock. But generic circuitry isn't enough. Language becomes universal because communicative input, unlike echo structure, inherits an orienting bias that makes its prediction errors developmentally consequential.

2.6 ANATOMICAL CONSIDERATIONS

Three anatomical traits are routinely cited as evidence of a language-specific adaptation: (i) the descended larynx, (ii) fine orofacial motor control, and (iii) cortical hemispheric lateralization for speech. Each, on closer inspection, is better read as an exaptation with its own adaptive history.

The descended larynx, long championed as a speech device, most plausibly evolved for acoustic size exaggeration in male competition, as indicated by sexual dimorphism and parallel descent in several loud-calling mammals (Fitch & Reby, 2001). Laryngeal descent occurs across multiple lineages during vocal display, so the trait is neither uniquely human nor necessarily language-driven (Fitch, 2010). For an opposing view see Lieberman (2017); Boë et al. (2019) provide a reply.

Primates possessed refined orofacial motor control well before language, primarily for feeding and facial expression (Burrows, 2008). Human elaboration likely reflects further selection for facial signalling and diet; articulate speech then co-opted this pre-existing control.

Hemispheric asymmetry is likewise not diagnostic of language. Functional lateralization for vocalization appears widely across vertebrates (Ocklenburg et al., 2013).

Songbirds and other vocal-learning taxa show left-dominant processing of species calls, confirming that lateralization is neither uniquely linguistic nor uniquely human.

These structures supply physical *scaffolding*. Language arises only when that scaffold is combined with three behavioural ingredients: species-universal vocal-production learning, infant-directed ostensive signalling from birth, and sensitivity to multi-timescale rhythm of the kind relevant to speech segmentation. Humans alone satisfy all three criteria in that strong form (Table 2).²

Table 2: Three attention-bias properties across vocal-learning taxa

Taxon (exemplar)	Vocal-production learning	Ostensive individually addressed signals	Sensitivity to multi-timescale rhythm / beat
<i>Homo sapiens</i>	Species-universal vocal-production learning (VPL) (Kuhl, 2004)	Gaze + prosody in infant-directed speech (IDS) (Vouloumanos & Werker, 2007)	Spontaneous beat entrainment (Large & Palmer, 2002)
Songbirds (zebra finch)	Juvenile males copy tutor song (Immelmann, 1969)	Directed courtship song; amplitude scales with distance (Brumm & Slater, 2006)	Discriminate isochronous vs. irregular; poor tempo generalization (van der Aa et al., 2015)
Parrots (African grey)	> 100 referential labels learned (Pepperberg, 1999)	Uses labels to address humans (Pepperberg, 1999)	Cross-tempo beat entrainment (Snowball) (Patel et al., 2009)
Bats (<i>Saccopteryx</i>)	Juveniles learn hover-song patterns (Knörnschild et al., 2010)	Hover-song directed at one harem female (Behr & von Helversen, 2004)	No beat entrainment reported (to my knowledge)
Cetaceans (bottlenose dolphin)	Calves invent signature whistles; adults copy others' whistles (Janik & Slater, 1998)	Whistle copying addresses a specific individual (King & Janik, 2013)	Turn-taking ≈ 1 s (Nakahara & Miyazaki, 2011); no beat entrainment
Pinnipeds (California sea lion)	Contextual vocal imitation (Reichmuth & Casey, 2014)	–	Head-bob entrainment to musical beats (Cook et al., 2013)

In sum, the descended larynx, orofacial dexterity, and hemispheric specialization each originated under pressures unrelated to speech and were later recruited for linguistic ends. Their convergence supports the view of language as an emergent capacity built on recycled anatomy rather than as a product of bespoke biological engineering.

3 THE ATTENTION-BIAS HYPOTHESIS

3.1 ATTENTION AS BIOLOGICAL GAIN CONTROL

The attention-bias hypothesis treats communicative salience as a neural “volume knob”, set over the bound event in which a vocalization is addressed and attended to, not over the auditory stream alone. In predictive-processing terms, the proposed knob sets the

²Fitch (2010, Ch. 14) proposes a melodic–rhythmic bias; the present account attributes the critical step to an attentional hyperprior that amplifies *any* multi-timescale rhythmic, ostensive event.

PRECISION of prediction errors across that event: high precision makes mismatches consequential for belief updating or action, whereas low precision lets them pass as noise.³

On this account, the boost is posited to operate over multimodal bindings of vocalization and context. Two cross-modal signatures are central to the human gain profile: (i) *ostensive cues*, including eye contact, infant-directed prosody, or motion; and (ii) *multi-timescale rhythm* whose envelope chunks into syllable- or gesture-sized units. Either cue partially up-weights precision; when both co-occur, as they typically do in infant-directed speech (IDS), the bound event is tagged “communicative” and routed through a high-gain channel.

Hypothesized sequence.

1. Cue detection: Subcortical detection circuits register ostensive and rhythmic cues, and cortico-basal-ganglia gating loops integrate them into a gain signal that tags the bound event as communicative. (These circuits are specialized for social/communicative gating, not for syntax or language per se.)
2. Precision boost: The tag raises expected precision across the cortical hierarchy and over cross-modal bindings.
3. Model updating: Amplified prediction errors drive rapid model adjustment.
4. Pattern stabilization: Recurrent regularities consolidate at high gain; comparable structure in low-gain channels goes uncaptured.

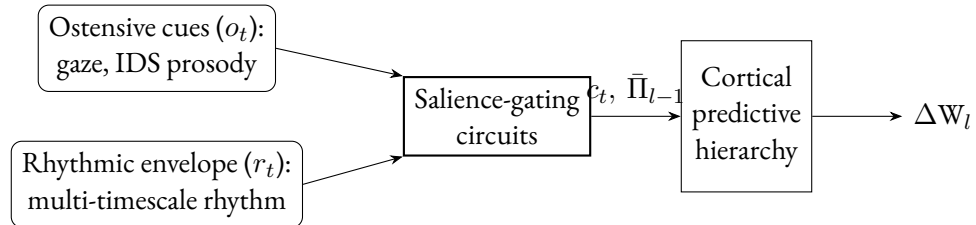


Figure 1: Hypothesized precision-weighting cascade. Salience-gating circuits (subcortical detection feeding cortico-basal-ganglia gating loops; §3.1) integrate ostensive (o_t) and rhythmic (r_t) salience signals into a gain-control signal (c_t), which raises expected precision ($\bar{\Pi}_{l-1}$) across cortical predictive layers and over cross-modal bindings. Amplified prediction errors drive synaptic updates (ΔW_l). The inherited specification is the policy that binds ostensive directedness with multi-timescale rhythm into a high-precision multimodal event channel; cortical learning is domain-general. Formal definitions in Appendix A.1.

If such a gain schedule is biologically implemented (Fig. 1), the inherited rule, *treat ostensive rhythmic events as reliable*, would steer a generic learner toward the linguistic

³Appendix A.1 formalizes the proposed gain schedule and learning-update rule; Appendix B reports a toy transitional-probability implementation that does not yield the predicted monotonic improvement. The formalization specifies the hypothesis; it does not validate it.

patterns of its community. On this account, the inherited contribution lies in sampling bias rather than in a dedicated syntax module; broader learning architecture and cultural input still do the representational work.

Gain here shouldn't be read as a homogeneous increase in arousal or as a single volume setting applied to the auditory stream alone. It's a precision-allocation policy over a bound multimodal event. The cue that triggers gain, the level at which gain is applied, and the learning system that receives the amplified error can dissociate. Ostensive cues raise the expected reliability of relations among vocal or signed form, rhythmic timing, speaker orientation, and the currently structured scene; the prediction errors that matter are relational, not only acoustic: mismatches between what's heard or seen, who's addressing whom, what's being attended to, and what the situation affords.

A learner may assign high precision to acoustic-template fidelity without comparable precision to phrase-level rhythm, distributional structure, or form–context binding. Nor does the account require ostension itself to be uniquely human: Gómez (2026) argues that great apes can display basic ostension through attention contact and behavioural utterances. The human-specific proposal therefore isn't that other taxa lack attentional gain or basic ostensive signalling, but that human infants inherit a distinctive gain profile: ostensive, multi-timescale, contextually framed input is treated as reliable evidence for general model updating across phonetic, syllabic, prosodic, contextual, and combinatorial scales.

Marblestone et al. (2016) sketch a compatible distinction. They argue that the brain optimizes cost functions which are themselves specified rather than learned, that those cost functions vary across brain regions and across development, and that this division keeps the genetic specification compact. Byrnes (2026) extends the picture into a volumetric two-subsystem view: a *Learning Subsystem* (cortex, hippocampus, amygdala, cerebellum; roughly 90% by volume) that learns from scratch, and a *Steering Subsystem* (hypothalamus, brainstem; roughly 10%) that houses genetically specified “business logic”: the cost functions, reward signals, and attentional priors that guide learning. The cortex, on this view, is a domain-general predictive engine; species-specific bootstrapping lives in the steering circuitry.

The division finds empirical support in single-cell atlases. Langlieb et al. (2023) mapped cell types across the adult mouse brain and found especially high cell-type diversity in the midbrain, hindbrain, and hypothalamus, precisely the steering regions, while cortical areas displayed a more uniform, repeating architecture. If evolution encodes complexity into loss functions rather than learning algorithms, the genome can stay compact: specifying a six-layer cortex takes little more code than specifying a three-layer one, but each bespoke reward heuristic demands its own wiring (cf. Marblestone et al., 2016). Computational modelling offers a proof of concept: Ullman et al. (2012)

showed that a single innate “mover” detector, an attentional prior for active, causal motion, was sufficient to bootstrap unsupervised learning of hands and gaze direction, concepts statistically inconspicuous but socially significant; state-of-the-art saliency and object-detection methods failed on the same input where the mover bias succeeded. The attention-bias hypothesis fits this picture if the inherited hyperprior for ostensive, multi-timescale communicative events is treated as a steering-subsystem cost function rather than as a cortical module. I read “steering subsystem” broadly: alongside the hypothalamic and brainstem nuclei in Byrnes’s narrower volumetric division, the basal ganglia and the cortico-basal-ganglia gating loops are the genetically-specified circuitry through which such a cost function would shape cortical learning. The cortex–basal-ganglia–thalamus loop is the vertebrate brain’s dopaminergic reinforcement-learning substrate; locating the inherited gain policy at the basal-ganglia–cortex (BG–cortex) interface keeps the steering/learning division compact while putting the inherited specification where the HAQER enrichment near striatal medium spiny neurons (§5) actually points.

3.2 DISTINGUISHING THE ATTENTION-BIAS HYPOTHESIS

Usage-based accounts and Schmidt’s *Noticing* hypothesis assign the bottleneck to conscious awareness or to the learner’s pattern-finding machinery (Schmidt, 1990). Tomasello’s social-cognitive model adds intention-reading as a second prerequisite (Tomasello, 2003). By contrast, the attention-bias proposal places the genetic change one level earlier: in a hyperprior, or small set of hyperpriors, that up-weights the bound event of ostensive, multi-timescale communicative input.⁴ The same explanatory target is reached without requiring intention-reading: infants inherit a precision profile that tags addressed communicative events as high-precision evidence for general model updating. The proposal keeps pattern-finding and theory-of-mind domain-general; what’s inherited is a precision filter on the bound event that determines which data enter those mechanisms with enhanced effective weight.

Deacon’s account of the symbolic leap (Deacon, 1997) requires sustained focus on linguistic form rather than on literal reference. A built-in precision boost for communicative events supplies that focus automatically, accomplishing the same explanatory work without positing new analytic machinery.

Laboratory work supports the distinction. Infants segment artificial grammars from non-social audio, but retention and generalization rise sharply when the same material is framed ostensively (Kuhl et al., 2003; Saffran et al., 1996). Social context seems to raise precision on the relevant prediction errors.

⁴Following Scott-Phillips (2015), I use OSTENSIVE CUE to mean any signal explicitly directed at the learner; here, though, ostension functions as an attentional prior, not as a full-blown pragmatic inference.

One might suggest that caregiver behaviour alone explains human language acquisition: in naturalistic field data, infants hear orders of magnitude more infant-directed vocalizations than any ape peer (roughly $70\text{--}400\times$ depending on the species; Wegdell et al., 2025). But disparity of input only restates the puzzle: what disposes human caregivers to supply, and human infants to draw, such intensive directed speech? An attention-bias account treats the caregiver side as a co-evolved complement to infants' attention: abundant directed input is necessary for fast acquisition, but on this account it becomes especially effective because infants are hypothesized to assign high precision to ostensive, multi-timescale rhythmic events. Without an attentional bias that makes those events disproportionately effective for general learning, even saturation training fails to yield language-like behaviour, as decades of ape tutoring attest (Terrace, 1979).

3.3 EVIDENCE FOR INNATE ATTENTIONAL BIASES

Comparative work shows that many species enter the world with built-in propensities to notice some stimuli over others. These biases channel information uptake and differ not only across taxa but even across strains within a species.

Field-cricket females, for example, orient toward a narrow frequency band that matches the conspecific calling song (~ 4.8 kHz); ascending interneuron 1 (AN1) is tuned to the same band, confirming a hard-wired mate-recognition filter (Wohlers & Huber, 1982). Domesticated foxes, selectively bred for sociability, detect human gaze direction far better than wild conspecifics (Hare et al., 2005). Snake-naïve Japanese macaques spot snake images faster than flower images, demonstrating an inherited bias toward ancestral threats (Shibasaki & Kawai, 2009).

White-crowned sparrow forebrain neurons start broadly tuned but shift their best frequency to the bird's own emerging song during early practice, a precision boost on the "self-produced motif" channel without peripheral change (Volman, 1993). Social context matters: juvenile zebra finches ignore tape playback but copy the same song when a live tutor is present, an effect interpreted as heightened attentional arousal rather than improved sensory fidelity (Doupe & Kuhl, 1999).

Bank-swallow parents accept any chick until day 15–16; once chicks develop individual "signature calls," parents learn them rapidly and eject intruders, a gain switch on caller identity (Beecher et al., 1981). Such cases illustrate Seligman's *preparedness*: some cue–outcome pairings are pre-weighted for rapid learning (Seligman, 1970).

Human newborns already show elevated responses to speech over acoustically matched non-speech (Vouloumanos & Werker, 2007). The bias initially extends to non-human primate vocalizations as well; species-specific tuning emerges over the first three months (Vouloumanos et al., 2010). By eight months they segment words using transitional probabilities (Saffran et al., 1996), but between six and ten months their discrimina-

tion of non-native phonemes declines, indicating a developmental re-weighting parallel to the animal data (Werker & Tees, 1984). Newborns also orient to face-like patterns with “eyes” above a “mouth” even under tight visual control (Farroni et al., 2005; Johnson et al., 1991), signalling high initial precision for the abstract structure of faces.

These findings show that evolution tunes the magnitude, target, and timing of attentional gain without altering basic sensory hardware. The same logic underpins the present hyperprior account: humans inherit high initial precision for ostensive, multi-timescale communicative events, ensuring that addressed input carries maximal weight in early learning.

3.4 ATTENTION VERSUS PROCESSING ACCOUNTS

Standard acquisition theories posit specialized machinery (Merge, constraint templates, parameter setting) to bootstrap grammar from scant input. The PoS argument makes that machinery indispensable. The attention-bias view shifts the locus of innateness: selection shaped a gain function that assigns high precision to ostensive, multi-timescale communicative events. Amplified prediction errors then drive a generic learner toward linguistic structure without language-specific algorithms.

Apparent support for processing solutions often dissolves under an attentional lens. Mismatch responses to native–non-native contrasts in seven-month-olds (Friederici, 2005) need not imply built-in phonological categories; infants could reach that discrimination via months of precision-boosted exposure. Likewise, laboratory segmentation of non-social sound strings is fragile, but performance soars when identical material is presented ostensively (Kuhl et al., 2003; Saffran et al., 1996).

Second-language evidence fits the same pattern. Adult second-language (L2) learners master high-salience vocabulary but struggle with low-salience morphology, category systems, and phonetic timing (Ellis, 2006; Schmidt, 1990). An early bias raises precision on those features for infants; adults have to re-engage it explicitly. Perceptual narrowing after 8–10 months (Werker & Tees, 1984) marks the point where stable predictions begin suppressing non-native contrasts.

Clinical dissociations sharpen the case. Developmental language disorder (DLD) impairs mass–count morphology but spares pragmatic article choice; autism spectrum disorder (ASD) shows the mirror profile (Creemers & Schaeffer, 2015). Eye-tracking confirms that DLD children still favour faces, where autistic peers don’t (Vacas et al., 2024). Resting-state EEG and tractography add a matching double contrast (Campos et al., 2025; Li et al., 2024), and a genome-wide association study (GWAS) reports near-zero genetic correlation between DLD and ASD (Nudel et al., 2024). These dissociations are consistent with separating domain-general auditory-motor constraints from atypical weighting of social cues, though neither clinical category maps cleanly onto a

single mechanism.

Adult L2 patterns and the DLD/ASD split together suggest that attentional weighting plays a substantial role alongside generic processing limits. Explicit instruction helps adults only when it shifts attention to low-salience features (Ellis, 2006). Minimal generic limits (oscillatory timing, shallow working memory) still bound learning, but only once precision sets the stage.

A loose computational analogy is worth flagging: transformer architectures achieve strong language performance with attention-based mechanisms (Vaswani et al., 2017), though biological and computational “attention” are different operations and the parallel is suggestive rather than evidential.

Predictive-processing theory casts attentional biases as adaptive priors that define what counts as surprising (Friston et al., 2012). Humans appear to inherit a prior that vocal signals are statistically rich, with species-specific calibration emerging through experience. No hard-wired syntactic operator is needed; the inherited contribution is just a bias to notice. The bee waggle dance makes the same point from the receiver’s side. In bees, the gain profile binds directional motion, rhythmic timing, and recent foraging context into mutually constraining evidence about a remote food source. Human observers see the same physical event but lack the precision policy that performs the binding; the dance stayed opaque until von Frisch dedicated enough general attention to notice the pattern (von Frisch, 1967). Without the right gain profile, structure can be present without being usable.

3.5 COMPARATIVE IMPLICATIONS: OTHER VOCAL LEARNERS

If an attentional hyperprior is the pivotal human innovation, why don’t songbirds, parrots, bats, and cetaceans evolve language-like systems? The answer can’t be that they lack gain. Vocal learners often show strong attentional, social, and reward-mediated amplification of particular sounds. The relevant variable is the *gain profile*: which cues trigger precision, which levels and bindings of the signal receive it, and which learning systems are updated by the resulting errors. Humans appear to assign high precision to ostensibly framed, multi-timescale communicative events: bindings of vocal or signed form, rhythmic timing, and addressed scene treated as evidence for general structural updating. Other vocal learners often assign high precision to different targets: tutor-song fidelity, conspecific-call identity, signature whistles, acoustic-template matching, or ambient sound reproduction. Those targets can support impressive learning without privileging the form–context coupling that supports open-ended combinatorial uptake.

Direct cross-species developmental evidence supports this asymmetry. Vouloumanos et al. (2010) found that human neonates show no preference for human speech over rhesus vocalizations, both preferred over synthetic sounds, but by three months human

infants strongly prefer speech. Parr et al. (2016) document the opposite trajectory in infant rhesus macaques: at birth they show conspecific-face preference that broadens with social experience. Ferry et al. (2013) report a related functional asymmetry. At three to four months, lemur calls support infant object categorization just as human speech does; by six months the advantage disappears for lemur vocalizations while persisting for speech, with the neural correlate (Woodruff Carr et al., 2021) showing modulated 4–9 Hz activity that sustains the language-cognition link for IDS but severs it for non-conspecific calls. The pattern is consistent with a developmental gain-shift: precision tagging of conspecific vocalizations narrows over the first six months in humans, while non-human primate infants traverse the analogous window in the opposite direction.

Songbirds illustrate the conserved mechanism. Chen et al. (2016) demonstrated that live social tutoring dramatically outperforms equivalent passive tape exposure in juvenile zebra finches, mediated by attentional gain: trial-by-trial variation in attention to the tutor predicts learning outcomes; tutors modify their songs into a directed register with infant-directed-like properties; and social interactions activate noradrenergic and dopaminergic midbrain neurons. Pilgeram et al. (2023) provided a direct pharmacological test: oxytocin-receptor antagonism during early vocal learning reduced attention to the tutor and adult song imitation accuracy, with early preference strength predicting later imitation fidelity. Zebra finches still track frequency contours with millisecond accuracy but show scant evidence of attending to supra-syllabic organization (Doupe & Kuhl, 1999); the basal ganglia implement a sophisticated reinforcement-learning loop in which dopaminergic signals bias exploratory motor variability toward better template matches (Fee & Goldberg, 2011), and the outcome of training is a fixed motif rather than open-ended combinatorics.

Open mimics are the strongest test case for this distinction. Their imitation is broad: parrots copy human speech sounds, mockingbirds copy songs of dozens of other species, and lyrebirds can reproduce striking environmental sounds. Such cases show that high acoustic gain isn't sufficient for language-like learning. The relevant asymmetry isn't *copying versus parsing*, since open mimics plainly segment, retain, and reproduce complex acoustic material. It's a contrast between precision on acoustic reproducibility and precision on the addressed event as a structured, world-directed episode. In humans, ostension raises gain on form–context relations: vocal or signed form, rhythmic timing, speaker orientation, joint attention, ongoing action, and the local scene are treated as mutually constraining evidence for model updating. The learner is not merely cued to hear a pattern, but to treat this form, from this agent, now, as bearing on this situation. Open mimics can assign high precision to the acoustic–motor match, and can reproduce acoustic patterns with extraordinary fidelity, without evidence for the same inherited bias to bind those patterns to event structure in a way that supports open-

ended communicative uptake. The grey parrot Alex is therefore best treated as a limiting case rather than as a mere surface mimic: intensive model–rival training yielded impressive referential and categorical learning, but didn’t produce a self-extending, population-sustained combinatorial system (Pepperberg, 1999). Broad imitation demonstrates acoustic gain; it doesn’t by itself demonstrate the human form–context gain profile.

For cetaceans, comparative evidence is thinner. Bottlenose dolphins can map acoustic tokens to objects or gestures, but gaining a few dozen symbols typically takes hundreds of reinforced trials per token and never yields spontaneous combinatorial use (Janik & Slater, 1997). Adult sperm whale escorts modify coda duration and diversity in the presence of calves (Baumgartner, 2025), an IDS-like signal modification on the production side. But no published study has tested matched neonatal preference for IDS-like versus non-IDS stimuli in any cetacean species; the developmental-attention side of the picture is unmeasured, and more broadly no published study has directly compared human and non-human neonates on the same matched IDS-vs-ADS preference paradigm in any species. The §4.1 cross-species design proposed below is an attempt to close that gap.

What’s species-specific in humans isn’t the existence of attention-driven gain, or even broad imitation, but the targeting of gain at the form–context coupling of communicative events: the multi-timescale rhythmic envelope binds with ostensive framing into a privileged channel for general model updating. Vocal learning, social attention, rhythmic sensitivity, and reinforcement-driven imitation are each widespread in partial form; what appears distinctive in humans is the strength, target, and developmental timing of the gain assigned to communicative input, which makes that input unusually effective in shaping general learning. The prediction is gradient: to the extent that a parrot, dolphin, or songbird received comparable endogenous gain on the form–context coupling of communicative events, the account predicts better language-like learning, bounded by species-specific differences in motor control, developmental timing, perceptual ecology, and the absence of a surrounding population already producing richly structured language. The Pilgeram et al. (2023) oxytocin manipulation in zebra finches is a partial empirical test in this direction: attenuating the gain reduces vocal learning, exactly as predicted. The §4.1 cross-species design extends this logic at the receiver side.

3.6 CAUSAL MODEL

Figure 2 makes the modelling assumptions explicit. G represents genetic variation relevant to attention, C the quantity and quality of caregiver input, A the genetically specified attentional hyperprior (strictly inherited; all environment-induced attentional tuning flows into L), L the learner’s internal trajectory (e.g. precision-weighted prediction-error dynamics), and Y the eventual language outcome. The model is scoped to species-

typical emergence in neurotypical development, holding sensorimotor capacity (oscillatory timing, working memory, vocal-motor control) fixed; individual variation in those capacities would require additional $G \rightarrow L$ or $G \rightarrow Y$ paths.

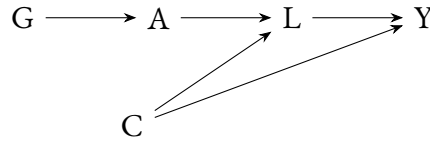


Figure 2: Core directed acyclic graph for the attention-bias account. G , genetics; C , caregiver input; A , genetically specified attentional hyperprior (the precision profile that binds ostensive, multi-timescale communicative events); L , learner trajectory (precision-weighted prediction-error dynamics); Y , language outcome. Caregiver input affects outcomes directly and via learning dynamics; in the simplified model A is strictly inherited (no $C \rightarrow A$ arrow).

In this simplified graph, A is exogenous with respect to C : caregiver input affects Y (directly and via L) but does not confound the causal effect of A on Y . The purpose of including C is not adjustment but bookkeeping: it records that language outcomes depend on the input environment, and it motivates the experimental strategy in Section 4, where C is held fixed or counterbalanced while the behavioural/neural signatures of A are assayed.

In the biological story, infants’ inherited A can of course *evoke* richer caregiver input over development (an $A \rightarrow C$ pathway), but that feedback loop is deliberately omitted from Fig. 2 because the proposed tests focus on birth-onset signatures and short-horizon assays rather than longitudinal co-regulation. The same logic applies within events: A drives orienting actions that shape which slice of C the infant samples in real time. This within-event feedback is also omitted from Fig. 2 for the same reason.

A fuller observational model would have to consider possible gene–environment correlation (e.g. parental genotype influencing caregiver input), but the tests proposed here avoid that complication by design (Pearl, 2009).

3.7 SCOPE OF THE MECHANISM: SEGMENTATION, STRUCTURE, AND WHAT IS PRE-SUPPOSED

The proposed mechanism, precision-weighting on the bound event, accelerates a delimited set of learning operations: segmentation, chunking, and pattern stabilization. Detecting an ostensive or prosodic communicative cue tags the bound event as high-value input, precision rises on the resulting prediction errors, and recurrent regularities are more likely to consolidate. The clearest experimental anchor is the dissociation reported by Thiessen et al. (2005), in which infant-directed and adult-directed prosody carried identical statistical structure, although only the IDS condition supported seg-

mentation in seven-month-olds. The gain operates on the same information; prosody changes its effective precision.

The proposed mechanism doesn't encode hierarchical syntactic structure in advance. A gain on prediction error changes the learning conditions under which structure can emerge: ostensive, multi-timescale events are treated as high-precision evidence, so recurrent regularities are more likely to be segmented, stabilized, generalized, and recruited into increasingly abstract patterns. The account doesn't posit a language-specific representational format or a syntax-specialized learning device, and stays neutral on whether the resulting structure is best modelled as a construction, a usage-based network, or a model-theoretic constraint system. Whatever it is, it's induced by general learning over high-precision input rather than pre-encoded by selection. How the brain implements this learning is a further question on which the account also stays neutral. The relevant computational test, when run as specified in §4.2, varies the precision profile while holding architecture constant.

Two contemporary neurolinguistic positions illustrate the shape of the open question. Friederici (2020) locates the Merge computation in posterior Broca's area, within a fronto-temporal network connected by the dorsal arcuate-longitudinal pathway whose late maturation is treated as the human-specific substrate for hierarchy. Fedorenko et al. (2020) report fMRI evidence that no region within the language network responds more strongly to syntactic than lexico-semantic processing, even when voxel subsets are defined by that contrast on half the data and tested on the other half. They argue that structural processing is spatially inseparable from processing word meanings.

The present account fits more naturally with the second picture than the first. Selecting for what we notice rather than for what we compute is consistent with structure being induced over a distributed substrate by gain signals, not extracted by a pre-specified hierarchy engine. But the view doesn't strictly contradict the localized picture: a gain on prediction errors during early development could in principle accelerate maturation of dorsal connectivity or other developmental scaffolding. The substantive point is what's assumed: whatever neural systems end up doing the stabilizing aren't under direct, language-specific selection in the present account. Selection is on the gain that makes communicative input weighty enough for general learning to organize.

One complication deserves explicit engagement. Çetinçelik et al. (2023) report that ten-month-old infants' neural tracking of continuous speech isn't modulated by the speaker's eye gaze, even though syllable-rate tracking predicts later vocabulary. The same paper notes that IDS prosody does enhance tracking and that eye gaze affects other measures of speech processing; the pattern is channel-specific, not uniformly null. Ludusan et al. (2022) similarly show that automatically-detected prosodic cues, unlike expert annotations, can fail to help and sometimes hurt some segmentation algorithms.

The bias account predicts that precision-weighting helps to the extent the gated channel is reliably detectable; such qualifications are best treated as constraints on cue specificity rather than evidence against the mechanism.

4 TESTING THE ATTENTION-BIAS HYPOTHESIS

The hypothesis yields three primary falsifiable predictions: (i) a communicative-tagging gain is present at birth in every human population, with an ostension-by-rhythm interaction in the §4.1 factorial (maximal response when both factors co-occur); (ii) the gain is absent or strongly attenuated in non-human comparison taxa lacking the human form–context gain profile, with a fuller comparative test in vocal-learning species deferred as a separate question; (iii) the gain, combined with rich communicative input, materially improves sample efficiency and learning trajectories in a generic neural architecture. Three complementary tests address these points.⁵

4.1 BIRTH-ONSET CROSS-SPECIES PREFERENCE TEST

Species: Human neonates (≤ 48 h), macaque neonates, and ferret kits (altricial; natural eye-opening at $\sim P_{30-35}$; widely used in developmental auditory neuroscience).

Stimuli: Four amplitude- and spectrum-matched tracks filtered to each species' audiogram, instantiating a 2×2 factorial of ostension $\times$ multi-timescale rhythm: (A) infant-directed speech (IDS; high ostension, high rhythm); (B) ostensibly framed non-rhythmic speech (high ostension, low rhythm; e.g., familiar maternal voice with the multi-timescale rhythmic envelope acoustically flattened); (C) modulation-matched rhythmic non-speech (low ostension, high rhythm); (D) non-rhythmic acoustic control matched on long-term spectral envelope (low ostension, low rhythm). Silence serves as a separate baseline characterizing neutral state.

Measure: Heart-rate deceleration (cross-taxon core measure) and, in humans only, pupil dilation.

Supportive outcome: In humans before 48 h, the precision-weighting prediction is an ostension-by-rhythm interaction (the factorial signature of form–context binding). A (IDS) should produce a response premium beyond what B (ostensive non-rhythmic) and C (rhythmic non-speech) jointly predict, with both intermediate cells exceeding D (non-rhythmic control). This pattern is hard to attribute to postnatal learning, though prenatal auditory exposure remains a confound to engage with directly. Failure to show it in the comparison taxa would be consistent with a human-specific communicative-

⁵The three tests play different roles with respect to *A* and *C*. In the behavioural preference and fNIRS studies, *A* is not manipulated; it's *assayed* by presenting tightly controlled stimuli (laboratory proxies for the cue-types that constitute *C* in vivo) and measuring the predicted ordering of responses. Only the computational replay explicitly *varies* an attention parameter (the gain profile) while holding architecture and training data fixed. Across all three designs, the point is to prevent input differences from masquerading as attentional differences.

tagging gain, subject to species-appropriate stimulus controls.

Disconfirming outcomes: three strict falsifiers, of increasing scope.

1. *Birth-onset version*. Across adequately powered human neonatal cohorts, with preregistered state exclusions and matched stimuli, no factorial cell elicits a response above the non-rhythmic acoustic control (D), or the Bayes factor favours the null across the factorial. This wouldn't refute all attention-bias accounts, but it would refute the strong birth-onset form defended here.
2. *Ostension $\times$ rhythm interaction*. The factorial reveals additive main effects rather than the predicted A-only premium. A and B equally elevated (ostension-only main effect) would support a social-pragmatic account; A and C equally elevated (rhythm-only main effect) would support a rhythm-first account; $A=B=C>D$ (any-stimulation main effect) would support an arousal account. Each of these patterns is empirically distinguishable from the precision-weighting interaction signature and would refute it.
3. *Human specificity*. Non-human neonates, tested with species-appropriate ostensive/rhythmic stimuli rather than merely human IDS proxies, show the same ordered gain profile.

The macaque and ferret arms function as a cross-taxon neonatal negative-control test rather than as a direct test of vocal-learning species; for these species the stimuli aren't assumed to be *conspecific*, the point being to test whether the human communicative-tagging gain generalizes to the acoustic-rhythmic proxy used here. The non-human arms run the three-cell subset (A, C, D); cell B's familiar-voice operationalization doesn't translate to non-human neonates without introducing a separate familiarity confound, so the full 2×2 interaction test is restricted to humans. The cross-species arms can usefully bound interpretation but shouldn't carry the main burden of prediction (ii); a stronger comparative test, in species-appropriate designs in vocal-learning taxa (song-birds, parrots, cetaceans), is a separate study.

Design: Sequential Bayes-factor stopping ($BF_{10} > 10$ or < 0.1) minimizes neonatal exposure. Parallel human cohorts (e.g., Toronto, Ho Chi Minh City) will check cross-cultural generality.

4.2 COMPUTATIONAL REPLAY WITH AND WITHOUT THE GAIN PROFILE

Capacity-matched learners receive the same multimodal input stream, but ostensive and multi-timescale rhythmic regions are tagged with higher sensory reliability (lower observation noise) in the human-like condition, with uniform reliability in the null condition, and with reliability shifted to non-communicative structure in the counterfactual. The manipulated parameter is a precision-allocation profile across input types, levels, and form-context bindings rather than a scalar intensity, consistent with the

structured-gain framing in §3.1. The manipulation operates on input precision before representation learning, not as a post-hoc loss-scaling term: scalar reweighting of the loss doesn't implement a hierarchical-Bayesian precision boost, since adaptive gradient normalization absorbs much of its effect. Metrics: speed to criterion and sample efficiency for word segmentation, subject-verb agreement, and hierarchical generalization probes (Warstadt et al., 2020). A learner under the human-like precision condition reaching criterion on the same input while the null condition lags would support prediction (iii): with architecture and input held constant, the precision manipulation materially shapes learning trajectories.⁶

4.3 NEONATAL NEURAL PRECISION-WEIGHTING

Quiet-alert neonates (12–24 h) undergo functional near-infrared spectroscopy (fNIRS) over bilateral superior temporal cortex. Conditions instantiate a graded form-context contrast: IDS (addressee-directed prosody with ostensive framing), adult-directed speech (ADS; third-person prosody, no addressee framing), and phase-scrambled speech matched for modulation spectrum (acoustic structure without communicative context). Target sample: 24 usable datasets (within-subject contrast, $d \approx .60$; $\sim 55\%$ attrition $\rightarrow$ 54 recruits).⁷ Precision metric: block-wise oxy-hemoglobin (HbO) amplitude (peak and 5–15 s area-under-curve) over bilateral superior temporal gyrus (STG) channels, with a pre-registered ordered-trend contrast IDS > ADS > scramble and a right-lateralization index for the IDS contrast. Newborn temporal cortex is differentially sensitive to the slow temporal structure that distinguishes these conditions (Telkemeyer et al., 2009); the prediction here is that this sensitivity is graded by communicative tagging.

The contrast tests a graded preference and its hemispheric organization, not a direct read-out of communicative-intent recognition; convergent controls dissociating IDS prosody from ostensive context would strengthen the inferential link by isolating the form-context binding from prosody alone.

Together, the preference test (prediction i + ii), the in-silico replay (iii), and the neonatal fNIRS assay (i) offer a minimal, directly falsifiable evaluation of the attention-bias account. Positive results would support an inherited hyperprior for addressed communicative events as the key evolutionary innovation behind human language, against

⁶A loosely parallel result is suggestive: Lee et al. (2026) report that transformers pre-trained on sequences from neural cellular automata converge faster on language and score higher on reasoning benchmarks than models pre-trained on natural text alone, with transfer concentrated in attention layers. The parallel is loose (Lee et al. vary training *data* rather than input reliability, and biological vs. computational “attention” are different operations), but it points toward structural priming of representational capacity prior to linguistic exposure as a route worth taking seriously.

⁷Planning assumption: the study powers for a moderate within-subject contrast ($d \approx .60$) and reports a sensitivity analysis over $d = .50$ to $.80$; attrition estimates should be taken from closely matched neonatal fNIRS protocols rather than from adult or older-infant studies.

alternatives that posit specialized syntax circuitry; they would not by themselves exclude every specialized-circuitry account.

4.4 TOY STRESS TEST OF A SINGLE-THRESHOLD TRANSITIONAL-PROBABILITY SEGMENTER

The Appendix B simulation, run as specified, does not bear out the monotonic prediction. Increasing the precision multiplier g from 1.0 to 2.0 produces a small monotonic decline in segmentation F1 (.729 to .716 across 10,000 paired trials, $\sigma \approx .004$) and a rising false-boundary rate on the adversarial distractor (.000 to .013). A sensitivity sweep over g , decision threshold, and distractor type shows that the gain is inert when the distractor is orthogonal or absent, and consequential mainly under contamination, where it shifts the precision–recall trade-off rather than producing a strict improvement. The simulation thus stress-tests a single-threshold transitional-probability (TP) segmenter; it does not corroborate the broader precision-weighting hypothesis. As Appendix B notes, the appropriate computational test of that hypothesis is an online learner with a structured precision profile and speed to criterion as the metric, since precision-weighting modulates learning dynamics rather than asymptotic estimates of conditional probability.

5 IMPLICATIONS FOR LANGUAGE EMERGENCE

An attentional hyperprior, not a bespoke syntax module, could be the decisive genetic tuning behind language. Selection would have adjusted the gain on ostensive, multi-timescale communicative events rather than inventing new processing machinery, a simpler trajectory within gene–culture coevolution.

Byrnes’s “thought assessor” mechanism sketches how a compact genetic specification can yield a richly generalized reward function (Byrnes, 2026). The steering subsystem contains innate heuristics (say, a flinch response when something small and dark skitters across the visual field). Part of the amygdala learns to *predict* that steering-subsystem response; as the predictor trains, it comes to fire not only when visual cortex detects actual spiders but also when the cortex entertains the *concept* of a spider or hears the *word spider*. Evolution doesn’t need to anticipate every trigger; it only specifies the innate heuristic and a slot for a predictor. The predictor inherits the cortex’s generalization.

The same architecture applies to ostensive cues. The steering subsystem can encode primitive detectors for eye contact and for rhythmic, pitch-modulated vocalization (the latter encountered even in utero; DeCasper and Fifer, 1980); both are features infants orient to within hours of birth. A cortical predictor learns to anticipate when those detectors fire. Once trained, the predictor generalizes: novel ostensive contexts (a pointing gesture, a vocative intonation contour, a raised eyebrow) activate the same high-

precision channel even though evolution never saw them. Tomasello (2026) provides a useful fit here: human index-finger pointing is developmentally robust and tied to coordinating collaborative activity, making it an especially plausible target for this learned generalization of an ostensive tag. The infant's world model grows to include "communicative" as a grounded category, and that category keeps recruiting new instances as exposure accumulates. No explicit encoding of "language" is required; only the primitive tag and a slot for learning.

Building on the structured-gain framing of §3.1, the form–context gain profile unifies perception and production. In an active-inference formulation, perception and action are two routes to prediction-error reduction: the learner may update its model of the attended event, or it may change its own output so that sensory consequences better approximate the attended target. Action here includes more than vocal convergence. The same precision policy that biases perception also drives orienting (gaze direction, head-turning, attention-deployment) toward addressed events, so the infant actively samples high-precision evidence rather than merely receiving it. On this view, infant vocal convergence doesn't require a second inherited reward specification. If ostensive, multi-timescale communicative events are treated as high-precision evidence, then mismatches between babble and the attended vocal target become consequential errors in the vocal-motor loop. Babbling is thereby biased toward the communicative event for the same reason perception is: the bound event has been assigned high precision. Songbird dopamine-gated tutoring (Tanaka et al., 2018) can be read as a lineage-specific implementation of this broader logic, not as evidence that the human account needs a distinct imitation-reward module.

The reframing still permits dissociations. A child may orient to speech while showing weak vocal convergence if the downstream motor loop, policy precision, or action-perception coupling is atypical. Such a profile wouldn't imply a missing imitation module; it would indicate that the same precision-weighted communicative event isn't being effectively translated into active vocal correction. Neuroimaging designs that separate input precision from motor-loop convergence could test this distinction.

That tweak dovetails with cultural transmission. Even weak learning biases, once amplified iteratively, yield systematic structure (Kirby et al., 2008). If the inherited bias is merely "listen harder when addressed," culture supplies the rest. This aligns with calls for mechanistic accounts of biological preparedness within gene–culture coevolution (Arnon et al., 2025). Their framework identifies "biological reward systems" as a recurring theme but leaves the mechanism unspecified; the present account fills that slot with a structured, testable parameter: an inherited precision profile on the bound event of ostensive, multi-timescale communicative input.

Genetic evidence points in the same direction. Claims of a recent *FOXP2* sweep have

collapsed under broader sampling (Atkinson et al., 2018). Genome-wide scans show diffuse polygenic signals (Hernandez et al., 2011; Mountford et al., 2022), consistent with selection acting on attention, motivation, and memory rather than on language-specific circuitry.

Recent genomic work sharpens this point. Casten et al. (2026) report that HAQERs, mostly noncoding regulatory regions that evolved after the human–chimpanzee split but before the human–Neanderthal split, are associated with spoken-language-related phenotypes. In the Epidemiology of Specific Language Impairment (EpiSLI) cohort, HAQER polygenic scores predicted sentence repetition and related language factors but showed no comparable association with nonverbal intelligence quotient (IQ); related spoken-language associations appeared in SPARK (an autism cohort), the Adolescent Brain Cognitive Development (ABCD) study, and UK Biobank.

The signal is polygenic, regulatory, and ancient: closer to developmental tuning than to a single-gene or *FOXP2*-style language-faculty story. The HAQER study reports enrichment near chromatin-accessible regions in medium spiny neurons and *FOXP2*-expressing neurons, plus elevated HAQER-like sequence similarity in vocal-learning mammals (Casten et al., 2026): the kinds of regulatory systems an attentional-tuning account would predict. For the present model, the result mainly supports $G \rightarrow Y$, with hints about $G \rightarrow$ neurodevelopmental regulation; it doesn't establish $G \rightarrow A$, since the measured outcomes are language scores rather than neonatal attention or precision-weighting. The picture fits the self-domestication route sketched in §2.5: regulatory tuning of the subcortical and basal-ganglia circuits that gate salience and reward, not new coding machinery.

Two further features of the HAQER results bear on the slogan that selection targeted what we notice rather than what we compute. HAQER variants are enriched for prenatal brain-development effects and head-circumference loci, with cognitive benefits balanced against the obstetric-dilemma costs of larger neonatal heads; and individual single-nucleotide polymorphisms (SNPs) in HAQERs carry roughly 188 times the predictive power for language of variants elsewhere in the genome (Casten et al., 2026). Both push gently against a strict reading on which the inherited contribution is confined to subcortical gain. The account accommodates a modest cortical developmental tilt (earlier or richer maturation of the cortical circuitry that the gain policy routes input through) alongside the BG–cortex gating that does the architectural work. The thesis survives: what selection added still falls well short of a syntax module. But “selection on what we notice” should be read with the cortical developmental footprint of HAQER variants in view.

As a future genomic extension, the account invites a stepped mediation test, with a 12–24-month intermediate as a prerequisite for the more demanding neonatal ver-

sion. The intermediate: a HAQER-overlap component of a cognitive-performance polygenic score (CP-PGS) (Casten et al., 2026), entered alongside the non-HAQER (background) CP-PGS, would predict toddler attentional uptake of IDS (sustained attention to addressed speech, looking-time to IDS over ADS) in a retrofittable cohort that pairs genotype with such measures in the 12–24-month window. The strict version, in which the same score predicts neonatal precision-weighting (IDS-related HbO amplitude and right-lateralized STG response are the cleanest candidate mediators) and that precision-weighting mediates later spoken-language outcomes, is the deferred follow-up: replication-scale effect sizes outside the EpiSLI discovery sample ($\beta \approx 0.05$) imply large- N neonatal cohorts beyond what current studies have run. A null on the toddler intermediate is a preregistration commitment, weakening the case for the more expensive neonatal design rather than redirecting around it. This is offered as a future direction rather than part of the present empirical program in §4.

Comparative evidence supports this evolutionary trajectory. While nonhuman great ape infants receive minimal directed vocal input and have to rely on surrounding communication (Wegdell et al., 2025), human infants are immersed in ostensive cues from birth. This suggests that early hominins likely resembled other apes in their reliance on eavesdropping and observational learning, with the massive expansion of infant-directed communication co-evolving with language itself. The 70–400-fold range increase in directed vocal input between humans and our closest relatives (Wegdell et al., 2025) implicates a fundamental shift in attentional weighting rather than wholesale computational restructuring.

The same distinction bears on the familiar contrast between children’s sample efficiency and the data demands of large language models (Frank, 2023). Human learners don’t merely receive fewer linguistic tokens; they receive tokens embedded in high-precision addressed events. Ostensive cues raise gain on form–context relations: vocal or signed form, rhythmic timing, speaker orientation, joint attention, ongoing action, and local affordances are treated as mutually constraining evidence. A single addressed utterance can therefore support updates over segmentation, reference, event structure, speaker reliability, and communicative expectation at once. Text-only pretraining supplies vastly more tokens, but those tokens are typically detached from the situated episodes that would identify which prediction errors should drive which updates. On this view, human sample efficiency reflects not only smaller data requirements or richer innate priors, but a species-typical precision-allocation policy that makes sparse inputs unusually informative.

The proposal explains why language acquisition is universal while other sophisticated skills (e.g., echolocation, absolute pitch, formal mathematics) are rare. Humans appear to show this bias robustly; other taxa (songbirds, dolphins, even highly encul-

turated primates) lack evidence for the full configuration of form–context gain, vocal learning, and cultural transmission. The bias drives infants toward speech, then perceptual narrowing refines the mapping as experience accumulates (Werker & Tees, 1984). No extra genetic specification is required, matching Heyes’s broader claim that many human cognitive specialities are culturally scaffolded rather than hard-coded (Heyes, 2019).

6 CONCLUSION

Emergence, seen through an attentional lens, ceases to be a puzzle of bespoke syntax machinery and becomes the predictable outcome of biased perception. Selection added not a signal filter but a binding bias (§3.1): a tendency to treat addressed communicative events as evidence about a structured world. A modest, inherited gain on ostensive, multi-timescale communicative input can steer generic learning mechanisms toward grammar: the same precision policy that makes speech worth learning from also makes self-produced deviations from attended vocal targets worth correcting. Culture supplies the structural details. This simple tweak aligns with gene–culture coevolution theory, bypasses the need for a language-specific module, and explains why language is universal in humans while other complex skills aren’t.

The account yields clear, falsifiable predictions: a communicative-tagging gain at birth (maximal when ostension and multi-timescale rhythm co-occur), its absence or attenuation in non-human comparison taxa, and measurable learning-trajectory effects *in silico*. The neonatal cross-species preference test, the computational replay with and without the gain profile, and the neonatal fNIRS assay outlined above together form an empirical program that can support, constrain, or refute the proposal.

Nothing here denies the intricacy of language; it relocates that intricacy to the interplay of a structured attentional profile with plastic neural circuitry and cultural iteration. We sought language in circuitry but find it in salience.

A FORMAL FRAMEWORK FOR ATTENTIONAL PRECISION AND LANGUAGE-LEARNING INPUT

A.1 ATTENTION AS PRECISION-WEIGHTING

Predictive-processing framework. A hierarchical predictive model treats perception as minimising free energy across levels $l = 1 \dots L$ (raw spectrogram $\rightarrow$ phonetic features $\rightarrow$ syllables $\rightarrow$ syntax). Each level generates a prediction $\hat{s}_{l-1}$ for the activity in the layer beneath and receives a prediction-error vector

$$\varepsilon_{l-1} = s_{l-1} - \hat{s}_{l-1}.$$

The influence of that error on synaptic updates is scaled by a **PRECISION** matrix Π_{l-1} – the inverse of expected noise covariance.

Operationalisation. We operationalise **ATTENTION** as the allocation of expected precisions $\bar{\Pi}_{l-1}$ that set the prior expectation of Π_{l-1} before any data are observed (Feldman & Friston, 2010). (Formally, $\bar{\Pi}_{l-1}$ is the mean of a hyperprior distribution over Π_{l-1} ; we use the expectation throughout for notational simplicity.) High expected precision amplifies the contribution of a given error term; low precision down-weights it.⁸ Unlike spotlight metaphors, this mechanism alters gain at every level simultaneously: information identified as communicative is boosted throughout the cortical hierarchy. We distinguish the realised precision Π_{l-1} from its prior expectation $\bar{\Pi}_{l-1}$; in what follows, we work with $\bar{\Pi}_{l-1}$ as a tractable proxy for Π_{l-1} (i.e. we treat learning as driven by expected precision rather than by trial-specific precision fluctuations).

Precision-allocation pathway.

1. **Cue detection.** Subcortical and early auditory circuits emit two graded salience signals at timestep t : an ostensive salience $o_t \in [0, 1]$ (mutual gaze, IDS prosody, turn-taking contingency) and a rhythmic-structure salience $r_t \in [0, 1]$ (multi-timescale envelope regularity that supports chunking). The two signals are integrated by cortico-basal-ganglia gating loops (the inherited gain policy is taken to sit at this basal-ganglia–cortex (BG–cortex) interface; see §3.1) into a single gain-control signal

$$c_t = \min(1, \alpha_o o_t + \alpha_r r_t + \alpha_{or} o_t r_t),$$

with $\alpha_o, \alpha_r \in (0, 1)$ and $\alpha_{or} \geq 0$ chosen so that IDS (high o_t and r_t) yields maximal gain, while rhythmic non-speech (high r_t , low o_t) yields an intermediate boost.

⁸Some authors argue that precision-weighting is not coextensive with attention; see Bowman et al. (2013) for event-related-potential (ERP) evidence that purposeful selection of expected stimuli involves mechanisms beyond prediction precision. The identification here is heuristic, not definitional.

2. Cue-weighted blending. Species-specific expected precisions $\{\bar{\Pi}_{l-1}^{\text{comm}}\}$ blend with baseline $\{\bar{\Pi}_{l-1}^{\text{env}}\}$ in proportion to c_t :

$$\bar{\Pi}_{l-1}(t) = \text{diag}(c_t \bar{\Pi}_{l-1}^{\text{comm}} + (1 - c_t) \bar{\Pi}_{l-1}^{\text{env}})$$

(assuming diagonal precision for simplicity).

3. Weighted error propagation. Each layer transmits the scaled error $\tilde{\varepsilon}_{l-1} = \bar{\Pi}_{l-1}(t) \varepsilon_{l-1}$ upward.
4. Learning update.

$$\Delta \mathbf{W}_l \propto \tilde{\varepsilon}_{l-1} \frac{\partial \hat{s}_{l-1}}{\partial \mathbf{W}_l}.$$

The inherited component is the mapping from ostensive and rhythmic evidence to expected precision – the pairing of $(\bar{\Pi}^{\text{comm}}, \bar{\Pi}^{\text{env}})$ together with the cue-detection circuitry that transduces ostensive markers into o_t , multi-timescale temporal structure into r_t , and both into the gain-control signal c_t (including the mixing parameters $\alpha_o, \alpha_r, \alpha_{or}$). This constitutes an *attentional gain schedule* that biases human infants toward treating ostensive, temporally structured signals as high signal-to-noise events across the hierarchy, making earlier extraction of linguistic regularities more likely.

A note on the structure of “gain.” In this notation, c_t isn’t the whole gain mechanism. It’s the cue-driven mixing coefficient that shifts the learner toward a communicative precision regime. The content of that regime is specified by the expected-precision profile $\bar{\Pi}^{\text{comm}}$: which error streams, at which levels, and across which cross-modal bindings are treated as reliable. The relevant prediction errors include not only acoustic mismatches at a single level of the auditory hierarchy but relational mismatches across signal, scene, speaker, and addressee. Species may therefore differ either in the cues that raise c_t or in the structure of $\bar{\Pi}^{\text{comm}}$. The human-specific claim concerns both: ostension and multi-timescale rhythm jointly raise c_t , and the resulting precision profile amplifies errors over the bound event, supporting segmentation, rhythmic grouping, and structural generalization rather than only acoustic-template fidelity.

A.2 DEFINING LANGUAGE-LEARNING INPUT

We call a sensory signal Σ LANGUAGE-LEARNING INPUT for a learner L when it is temporally parsable and (optionally) marked as addressee-directed, with ostension modulating the *weight* assigned to parsable input. (Some dependence is expected: infant-directed speech both signals addressee-directedness and alters temporal structure.)

(C1) Ostensive cue index. Let $O_t \in [0, 1]$ be the posterior probability, output by a classifier trained on a video corpus of caregiver–infant and adult–adult interactions,

that the signal at frame t is addressed to a specific recipient.⁹ We define the indicator $\tilde{O}_t = \mathbb{1}[O_t > 0.7]$, a binary gate that selects frames likely to be addressee-directed. The indicator $\tilde{O}_t$ operationalises the presence of an OSTENSIVE CUE without requiring linguistic comprehension or intention attribution by the learner.

(C2) Parsability index. From a separate set of acoustic/visual statistics, we compute $P_t \in [0, 1]$ from the modulation spectrum of Σ : P_t rises when energy clusters around 2–10 Hz (syllabic/sign strokes) and 0.3–3 Hz (phrase/gesture groups) and falls when the envelope is aperiodic and approximately scale-free (Giraud & Poeppel, 2012). P_t quantifies how readily the raw stream can be chunked into nested or multi-scale temporal units by generic oscillatory mechanisms. ($\tilde{O}_t$ is a coarse behavioural proxy for the ostensive component o_t ; P_t is a proxy for rhythmic salience r_t . Together they provide an operational proxy for the gain control posited in §A.1; we do not assume that the learner explicitly computes W or that W recovers the internal variable c_t .)

A segment $\Sigma_{t_0:t_1}$ becomes LANGUAGE-LEARNING INPUT for L when the joint input-gating score

$$W(\Sigma_{t_0:t_1}) = \frac{1}{t_1 - t_0 + 1} \sum_{t=t_0}^{t_1} \left(w_R + (w_O - w_R) \tilde{O}_t \right) P_t$$

exceeds a species-typical threshold θ_W , where $0 < w_R < w_O$. This yields the qualitative ordering IDS > rhythmic non-speech > non-rhythmic acoustic control, with silence as a separate baseline (P_t high in the first two, low in the non-rhythmic control, and near-zero in silence; $\tilde{O}_t$ high only in IDS). Condition C1 depends only on low-level behavioural markers; C2 on acoustic/visual statistics. Neither presupposes that Σ is language – only that it is temporally structured, and that it may (or may not) display addressee-directed cues. When $W > \theta_W$, the attentional system treats prediction errors in that segment as high-precision, triggering the learning loop described in §A.1.

⁹Candidate corpora include the Infant-Directed Communication dataset (Databrary, 44 dyads) and the Shared Book Reading Corpus (44 dyads). Cross-cultural extension remains future work. Annotation uses directly observable features: gaze alignment, head orientation, contingency in turn-taking, and amplitude/pitch exaggeration for speech or sign.

B TOY TRANSITIONAL-PROBABILITY STRESS TEST OF A SINGLE-THRESHOLD SEGMENTER

The following simulation isolates the ostensive component of the weighting scheme. As specified, it tests whether a multiplicative precision boost g on ostensive-flagged tokens improves boundary detection in a single-threshold transitional-probability segmenter when the input is contaminated by an adversarial non-communicative distractor. The simulation is reported here for completeness; as the closing remarks make clear, this implementation is a stress test of a particular offline scoring procedure, not a test of the broader hypothesis advanced in §A.1.

Design. We generate a mixed stream from two sources:

1. Structured (language-like): bigram sequences drawn from a three-word lexicon (ab, cd, ef) with deterministic within-word transitions and approximately uniform cross-boundary transitions.
2. Structured distractor (non-ostensive): a non-communicative stream with strong local transitional probabilities (e.g. a repeating ba pattern), which contaminates bigram statistics in ways that can mask true word boundaries if treated as equally reliable input.

An “ostensive flag” variable (`ost` in code) is set to 1 for structured-source tokens and 0 for distractor tokens. The precision weight $w_t = g \cdot \text{ost}_t + (1 - \text{ost}_t)$, where $g \geq 1$ is the gain parameter. In the notation of §A.2, the toy sets $P_t \equiv 1$, treats `ost` as $\tilde{O}_t$, and corresponds to $w_R = 1$ and $w_O = g$.

Segmentation. A simple transitional-probability segmenter inserts a boundary wherever the weighted bigram probability drops below a fixed threshold. Performance on the structured portion is measured by F1 score against gold-standard word boundaries; for the distractor portion, we report the false-boundary rate.

Hypothesis (as tested). If the toy adequately models the bias, increasing g should raise F1 on the structured stream (the weighted statistics better reflect true word boundaries), and leave the distractor false-boundary rate stable or falling.

```
import random, collections, statistics

def gen_mixed_stream(n_struct=30, n_noise=20):
    words = [['a', 'b'], ['c', 'd'], ['e', 'f']]
    struct = []; gold = set()
    for _ in range(n_struct):
        w = random.choice(words)
```

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        struct += w
        gold.add(len(struct)-1)
        noise = (['b', 'a'] * ((n_noise + 1)//2))[:n_noise]
        stream = struct + noise
        ostensive = [1]*len(struct) + [0]*len(noise)
        return stream, gold, ostensive, len(struct)

def segment(s, w, thr=0.4):
    uni = collections.defaultdict(float)
    bi = collections.defaultdict(float)
    for i, x in enumerate(s):
        uni[x] += w[i]
        if i > 0:
            bi[(s[i-1], x)] += w[i]
    boundaries = set()
    for i in range(len(s)-1):
        pair = (s[i], s[i+1])
        if uni[s[i]] > 0 and bi[pair]/uni[s[i]] < thr:
            boundaries.add(i)
    return boundaries

def f1(pred, gold):
    if not pred and not gold: return 1.0
    if not pred or not gold: return 0.0
    tp = len(pred & gold)
    return 2*tp / (len(pred) + len(gold))

def run_one(seed, runs=500):
    random.seed(seed)
    # Pre-generate streams so every g value sees identical data (paired)
    streams = [gen_mixed_stream() for _ in range(runs)]
    out = {}
    for g in [1.0, 1.25, 1.5, 1.75, 2.0]:
        f_struct, noise_rate = 0, 0
        for s, gold, ost, split in streams:
            wt = [g if ost[i] else 1.0 for i in range(len(s))]
            pred = segment(s, wt)
            gold_s = {b for b in gold if b < split}

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        pred_s = {b for b in pred if b < split}
        f_struct += f1(pred_s, gold_s)
        pred_n = {b for b in pred if b >= split}
        noise_rate += len(pred_n) / (len(s) - split)
    out[g] = (f_struct/runs, noise_rate/runs)
    return out

all_runs = [run_one(s) for s in range(20)]
for g in [1.0, 1.25, 1.5, 1.75, 2.0]:
    f1s = [r[g][0] for r in all_runs]
    noises = [r[g][1] for r in all_runs]
    print(f"g={g}: F1 mean={statistics.mean(f1s):.4f} "
          f"sd={statistics.stdev(f1s):.4f} "
          f"noise mean={statistics.mean(noises):.4f} "
          f"sd={statistics.stdev(noises):.4f}")

```

The listing implements the paired-stream version: identical streams generated for each gain level so that comparisons are within-stream, across 20 fixed random seeds $\times$ 500 trials per g (10,000 paired trials per condition). A minimal single-pass version over three values of g produces the same qualitative pattern; the paired version is shown here so the displayed listing reproduces the reported variance estimates. The fuller distractor-type and threshold sweep is summarised at the end of this section.

Result. The simulation does not produce the predicted pattern. With the submitted parameters ($g \in \{1.0, 1.5, 2.0\}$, threshold 0.4, lexicon ab/cd/ef, adversarial distractor ba), F1 falls monotonically from .729 to .716 across 10,000 paired trials ($\sigma \approx .004$); the false-boundary rate on the distractor rises monotonically from .000 to .013. Table 3 reports the paired-stream sweep across 20 seeds $\times$ 500 trials per gain level. A full sweep over $g \in [1, 10]$, decision threshold $\in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, and distractor type (adversarial, orthogonal, random, none) shows that the gain is inert when the distractor is orthogonal or absent, and consequential mainly under contamination, where it shifts the precision–recall trade-off rather than producing a strict improvement. No regime gives both rising F1 and stable noise rate.

Interpretation. The result confirms a previously noted caveat: a single-threshold transitional-probability segmenter is non-monotonic in g when cross-boundary bigrams are artificially frequent (the chosen distractor ba is exactly that case). What the simulation thus illustrates is the unsuitability of single-threshold offline TP scoring as a test of precision-weighting, not a failure of the broader hypothesis. Under the form–context framing of

Table 3: Paired-stream sweep ($20 \text{ seeds} \times 500 \text{ trials per } g$). Both metrics are monotonic in the wrong direction relative to the prediction.

g	Structured F1 (mean $\pm$ standard deviation)	Noise-boundary rate (mean $\pm$ standard deviation)
1.00	0.7291 ± 0.0042	0.0000
1.25	0.7274 ± 0.0042	0.0000
1.50	0.7252 ± 0.0043	0.0005
1.75	0.7214 ± 0.0040	0.0047 ± 0.0020
2.00	0.7156 ± 0.0034	0.0133 ± 0.0032

§3.1 the mismatch is sharper still: scalar gain on signal-internal transitional probabilities tests only a degenerate one-dimensional slice of the structured precision profile across input types, levels, and form–context bindings that the account requires. The appropriate computational test would examine *speed to criterion* under an online learner with a structured precision profile, since precision-weighting modulates learning rate rather than asymptotic estimates of conditional probability.

C IDENTIFICATION OF THE CAUSAL EFFECT OF ATTENTION ON LANGUAGE

Relating this appendix to Fig. 2. Figure 2 in the main text is a short-horizon, birth-onset simplification used to motivate the proposed neonatal assays; it omits the longer-run evocative path by which an infant’s attentional phenotype can shape subsequent caregiver input. The present appendix reinstates that developmental feedback ($A \rightarrow C$) to clarify identification in observational data.

The target estimand is the *average causal effect* (ACE) of attentional bias on language outcome:

$$\text{ACE}_{A \rightarrow Y} = E[Y \mid \text{do}(A=1)] - E[Y \mid \text{do}(A=0)].$$

Interpreting A . In observational contexts, A denotes a measured attentional phenotype (e.g., orientation latency to ostensive cues). Caregiver input C is best treated as a *mediator* or *consequence* of A : infants who attend more strongly to social cues elicit richer infant-directed speech.

Confounding structure. Let U denote unmeasured confounders that affect *both* the child’s attentional phenotype and language outcomes via routes other than attention itself – including genetic factors, socio-economic status (SES), and prenatal health. U plausibly also affects caregiver input (e.g., SES shapes quantity and quality of IDS). The directed acyclic graph (DAG) is:

$$U \rightarrow A \rightarrow Y, \quad U \rightarrow Y, \quad U \rightarrow C, \quad A \rightarrow C \rightarrow Y.$$

Conditioning on C is inappropriate for the total effect: it blocks the mediated path $A \rightarrow C \rightarrow Y$ and, because U also affects C , can open the collider path $A \rightarrow C \leftarrow U \rightarrow Y$. We *do not* adjust for caregiver input when estimating the total effect $A \rightarrow Y$.

Experimental identification. In experimental designs, we randomise an *exposure* E (e.g., presence vs. absence of ostensive cues) and measure an immediate attentional/neural marker M (e.g., heart-rate deceleration, pupil dilation, functional near-infrared spectroscopy (fNIRS) oxy-hemoglobin (HbO) amplitude over superior temporal gyrus (STG) channels) that serves as a proxy for A . Randomisation makes $E \perp U$, so contrasts in M across conditions identify the causal effect $E \rightarrow M$; interpreting these assays as evidence about later language outcomes Y requires an additional (typically longitudinal) link from M to Y .

Observational estimation. For observational data, we estimate

$$Y = \beta_0 + \beta_1 A + \gamma^\top f(U_{\text{obs}}) + \varepsilon,$$

where $f(U_{\text{obs}})$ encodes observable proxies for U (SES, birth weight, parental education, etc.). The coefficient $\hat{\beta}_1$ estimates $\text{ACE}_{A \rightarrow Y}$ under the assumption that no unmeasured confounding path $U \rightarrow A, U \rightarrow Y$ remains open.

SENSITIVITY ANALYSIS

Because unmeasured confounding cannot be ruled out in observational studies, we recommend two sensitivity metrics:

1. *E-values* (VanderWeele & Ding, 2017): the minimum strength of an unmeasured confounder (on the risk-ratio scale) needed to nullify $\hat{\beta}_1$.
2. *Robustness values* based on partial R^2 (Cinelli & Hazlett, 2020): the proportion of residual variance that an omitted variable would need to explain in both A and Y to reduce $\hat{\beta}_1$ to zero.

These metrics quantify the robustness of the causal conclusion without positing specific additional variables.

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